

De-multiplexing of Address/Data Bus (Self-Learning)

Q. Why does the 8086 use a multiplexed Address/Data bus Explain the process of de-multiplexing this bus using Address Latch Enable (ALE) and Latches.

[DEF] Definition to Remember

To save physical pins on the 40-pin 8086 IC, Intel used a Multiplexed Bus (AD0-AD15), where the same pins carry the Address during clock cycle T1, and Data during T2-T4. De-multiplexing is the hardware process of using external Latches and the ALE signal to extract and hold the address before the pins switch to data mode.

1. Need for Multiplexing and De-multiplexing

* Multiplexing: The 8086 needs 20 pins for the address and 16 pins for data. Having 36 separate pins on a 40-pin chip leaves no room for power, clock, and control signals. Hence, they are multiplexed.

2. The Role of ALE (Address Latch Enable)

* The 8086 generates a positive pulse on the ALE pin strictly during the T1 state of every machine cycle.

3. Using Latches (74LS373) for De-multiplexing

To de-multiplex, external D-type transparent latches (e.g., 74LS373) are used.

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Multiplexing saves pins (AD0-AD15 act as Address in T1, Data in T2-T4).
2. Memory requires a stable address for the entire cycle, creating the need for De-multiplexing.
3. De-multiplexing separates the Address and Data using external Latches (e.g., 74LS373).
4. The ALE (Address Latch Enable) signal is pulsed High by the CPU only during T1.
5. ALE connects to the Enable pin of the external latches.
6. When ALE is High (T1), the latches grab the address from the AD bus.
7. When ALE goes Low, the latches lock the address and hold it steady for the memory, while the CPU uses the AD bus for data in T2-T4.

[RECALL] Quick Recall

Multiplexing saves pins -> Memory needs stable address -> Use Latches (74LS373) + ALE signal -> T1: ALE is High, Latches grab Address
-> T2-T4: ALE Low, Latches hold Address, Bus used for Data